

Physicochemical Analysis of Water Samples Sources from Wells and Boreholes in Residential Areas Close to Cemeteries: A Review

¹Hannatu Abidu Aliyu, ²Aminu Akilu, ²Hauwa'u Salmanu, ²Abubakar Lawal and ³Alkali Mohammed

¹School of Interim and Preliminary Studies, Federal University Dutsin-ma, Katsina State, Nigeria.

²Department of Pure and Industrial Chemistry, Umaru Musa Yar'adua University Katsina, Nigeria

³Department of Chemistry, Yobe State University, Damaturu, Nigeria

*Corresponding authors' email: hannatuabidu97@gmail.com Phone: +2348067960795

ABSTRACT

The proximity of groundwater sources to cemeteries poses a significant public health risk due to the potential leaching of contaminants from decomposing human remains. This review synthesizes literature on the physicochemical quality of water from wells and boreholes near Nigerian cemeteries, aiming to evaluate contamination levels against WHO and NSDWQ standards. Findings consistently reveal deteriorated water quality, with parameters such as pH (5.8–8.9), electrical conductivity (450–2800 $\mu\text{S}/\text{cm}$), total dissolved solids (210–1950 mg/L), chloride (85–650 mg/L), nitrate (5–120 mg/L), sulphate (40–400 mg/L), and heavy metals like lead (0.02–0.45 mg/L) and iron (0.35–2.10 mg/L) frequently exceeding permissible limits. Microbial contamination, particularly coliform bacteria, is also prevalent, indicating faecal pollution. The evidence confirms that cemeteries act as point sources of groundwater pollution, with contamination influenced by hydrogeological factors and burial practices. This review underscores the urgent need for stricter regulatory guidelines on cemetery siting, improved leachate management, and continuous water quality monitoring to safeguard public health. It calls for integrated environmental management and further research into remediation strategies for affected communities.

Received: 15 Dec. 2025

Accepted: 30 Dec. 2025

Published: 29 Sept. 2026

Keywords:

Groundwater contamination, Cemetery leachate, Physicochemical analysis, Water quality, Public health, Nigeria.

INTRODUCTION

Access to safe and potable water is a fundamental human right and a critical determinant of public health. In Nigeria, groundwater sourced from wells and boreholes remains a primary water supply for a vast majority of the population, particularly in suburban and rural residential areas (Akoto *et al.*, 2019). However, the safety of this groundwater is perpetually threatened by various anthropogenic activities, including improper waste disposal, agricultural runoff, and industrial effluent. One often-overlooked potential source of groundwater contamination is the cemetery.

Cemeteries are repositories for human corpses, which upon decomposition, release a complex leachate into the surrounding soil. This leachate is rich in organic matter, inorganic salts, heavy metals (from coffin materials and bodily implants), and pathogenic microorganisms (Ücisik & Rushbrook, 1998). The migration of this leachate plume through the subsurface strata can eventually pollute underlying aquifers, especially in areas with shallow water tables and permeable soils, which are common in many parts of Nigeria (Adewumi *et al.*, 2021). Consequently, residents relying on nearby wells and boreholes face potential exposure to harmful contaminants. This review synthesizes evidence from various studies to evaluate the impact of cemetery leachate on the physicochemical quality of groundwater in Nigerian residential areas. The primary objectives of this review are: to evaluate and compare the physicochemical parameters of water samples from wells and boreholes situated near cemeteries against both Nigerian (NSDWQ) and international (WHO) drinking water standards; to identify the key contaminants of concern and trace their potential sources within the cemetery environment, including decomposition by-products, coffin materials, and embalming fluids; and to

synthesize the collective findings to highlight the public health implications for dependent communities, ultimately providing evidence-based recommendations for regulatory enforcement and sustainable mitigation strategies.

Related Literature Review

Numerous studies across Nigeria have investigated the impact of cemeteries on groundwater quality. A study in Ikhueniro cemetery, Benin City, found that well water within 50 meters had significantly elevated levels of EC, TDS, Cl^- , NO_3^- , and total hardness compared to control samples from a farther distance (Obaseki & Ikenweije, 2017). Similarly, research around the Ayobo cemetery in Lagos reported that borehole and well water samples were contaminated with microbes and heavy metals, including Pb and Fe, attributing this to leachate infiltration (Adeyemi *et al.*, 2019).

In the vicinity of the Atan cemetery in Yaba, Lagos, investigations revealed that the mean values for TDS, COD, BOD, and phosphate in nearby wells exceeded WHO limits, indicating organic pollution from decomposing matter (Olafusi *et al.*, 2018). A comparative study in Ibadan by Sangodoyin and Amori (2019) showed that water samples from wells downstream of a cemetery had higher concentrations of ions (Ca^{2+} , Mg^{2+} , K^+) and anions (SO_4^{2-} , NO_3^-) than those upstream, confirming the cemetery's role as a pollution point source.

Studies in northern Nigeria echo these findings. Around the Tudun Wada cemetery in Kano, well water was found to be contaminated with high levels of coliform bacteria and physicochemical parameters that rendered it unsafe for consumption without treatment (Bala *et al.*, 2020). Research in Kaduna Metropolis concluded that the shallow aquifer near a major cemetery was highly vulnerable to pollution, with

trace metals showing increased concentration with proximity to the gravesites (Musa *et al.*, 2021).

Further studies in Abia State (Nnadozie *et al.*, 2020), Port Harcourt (Wami *et al.*, 2022), and Osun State (Akinbile *et al.*, 2021) all consistently report a pattern of deteriorating water quality—characterized by elevated ions, nutrients, and microbial loads—in groundwater sources adjacent to cemeteries. The consensus across these twelve studies is that cemetery leachate poses a tangible and measurable risk to

groundwater quality, with contamination levels influenced by factors such as soil type, depth to water table, aquifer permeability, and age of the cemetery.

The consensus across these studies is that cemetery leachate poses a tangible and measurable risk to groundwater quality, with contamination levels influenced by factors such as soil type, depth to water table, aquifer permeability, age of the cemetery, and the types of materials buried (coffins, embalming fluids).

Table 1: An Overview of the main Physicochemical Characteristics of Research on Water Sources Close to Nigerian Cemeteries

Parameter	WHO Standard (2011)	NSDWQ Standard (2015)	Range obtained	Remarks	Reference(s)
pH	6.5 - 8.5	6.5 - 8.5	5.8 - 8.9	Often slightly acidic to neutral, sometimes within range, but values outside range indicate contamination.	Adeyemi <i>et al.</i> , 2019; Olafusi <i>et al.</i> , 2018; Bala <i>et al.</i> , 2020
EC ($\mu\text{S}/\text{cm}$)	1000	1000	450 - 2800	Frequently exceeds standards, indicating high ionic content from leachate	Obaseki & Ikenweuwe, 2017; Adeyemi <i>et al.</i> , 2019; Musa <i>et al.</i> , 2021
TDS (mg/L)	500	500	210 - 1950	Often above permissible limits, indicating high mineral and salt content	Obaseki & Ikenweuwe, 2017; Olafusi <i>et al.</i> , 2018; Wami <i>et al.</i> , 2022
Cl^- (mg/L)	250	250	85 - 650	Elevated levels common, derived from decomposition fluids	Obaseki & Ikenweuwe, 2017; Sangodoyin & Amori, 2019; Nnadozie <i>et al.</i> , 2020
NO_3^- (mg/L)	50	50	5 - 120	Often exceeds limits in older cemeteries or areas with shallow water tables.	Sangodoyin & Amori, 2019; Akinbile <i>et al.</i> , 2021; Musa <i>et al.</i> , 2021
SO_4^{2-} (mg/L)	250	100	40 - 400	Frequently high, can exceed both WHO and stricter NSDWQ standards	Sangodoyin & Amori, 2019; Adewumi <i>et al.</i> , 2021
Total Hardness ($\text{mg}/\text{L CaCO}_3$)	500	500	150 - 850	Often elevated due to dissolution of calcium and magnesium from bone matter and soil.	Obaseki & Ikenweuwe, 2017; Bala <i>et al.</i> , 2020
BOD (mg/L)	-	-	5.2 - 18.7	High values indicate significant organic pollution from decomposing matter.	Olafusi <i>et al.</i> , 2018; Wami <i>et al.</i> , 2022
COD (mg/L)	-	-	12.8 - 48.3	Confirms the presence of oxidizable organic material.	Olafusi <i>et al.</i> , 2018; Akinbile <i>et al.</i> , 2021
Lead (Pb) (mg/L)	0.01	0.01	0.02 - 0.45	Often detected above limits; linked to coffin materials and artefacts.	Adeyemi <i>et al.</i> , 2019; Musa <i>et al.</i> , 2021
Iron (Fe) (mg/L)	0.3	0.3	0.35 - 2.10	Frequently exceeds standards; can cause aesthetic and health issues.	Adeyemi <i>et al.</i> , 2019; Bala <i>et al.</i> , 2020
Total Coliform (CFU/100ml)	0	0	Consistently present, often high (>100)	Universal indicator of faecal and decomposition contamination.	Bala <i>et al.</i> , 2020; Wami <i>et al.</i> , 2022; Nnadozie <i>et al.</i> , 2020

Discussion

Cemeteries serve as multi-contaminant point sources, significantly changing the physicochemical and microbiological integrity of nearby groundwater systems, as demonstrated by the combined evidence from the reviewed research. The persistent departure from WHO (2011) and NSDWQ (2015) thresholds in Nigeria's various geopolitical zones is a predictable geochemical reaction to the strong biogeochemical transformation of buried organic matter rather than an accident.

The pronounced elevation in Electrical Conductivity (EC) and Total Dissolved Solids (TDS), frequently exceeding 1000 $\mu\text{S}/\text{cm}$ and 500 mg/L respectively (Obaseki & Ikenweuwe, 2017; Wami *et al.*, 2022), reflects the massive ionic flux generated during the mineralization of soft tissues. As cellular

macromolecules—proteins, lipids, and nucleic acids—undergo hydrolytic cleavage, they liberate a complex cocktail of soluble inorganic salts. Specifically, the elevated chloride (Cl^-) concentrations (up to 650 mg/L) are diagnostic of the high chloride content in bodily fluids and blood plasma, while the notable increases in nitrate (NO_3^-) and sulphate (SO_4^{2-}) provide critical insights into the decomposition pathway. Elevated nitrate, particularly pronounced in older cemeteries (Sangodoyin & Amori, 2019; Akinbile *et al.*, 2021), is indicative of the complete aerobic nitrification of ammoniacal nitrogen—a major end-product of protein deamination. Conversely, the presence of elevated sulphate (exceeding 400 mg/L in some cases) is directly attributable to the oxidative breakdown of sulfur-bearing amino acids (cysteine and

methionine) prevalent in keratinized tissues and skeletal muscle (Adewumi et al., 2021).

In addition to the principal ions, the regular identification of hazardous heavy metals, especially iron (Fe) and lead (Pb), exceeding allowable levels (Adeyemi et al., 2019; Musa et al., 2021) calls for further investigation. Unlike nit nit nit nit nit nit nit nit nit nit nit nit nit nit nit nit. They are mostly caused by the corrosion of non-biodegradable burial artifacts, such as painted surfaces, metallic coffin handles, galvanized iron nails, and, historically, lead-based soldering or embalming substances. Because of their bioaccumulative toxicity and resistance to natural attenuation, the persistence of heavy metals in groundwater, even at low levels ($Pb \geq 0.01$ mg/L), is especially problematic.

The microbial dimension further compounds the chemical hazards. The universal prevalence of total coliform bacteria in well water proximal to cemeteries (Bala et al., 2020; Wami et al., 2022) is arguably the most immediate health red flag, as it signifies direct faecal-organic contamination pathways. Unlike chemical pollutants that may be partially adsorbed onto soil colloids, pathogenic bacteria and viruses exhibit remarkable hydraulic mobility in porous media, especially during storm events when water tables rise rapidly. The significant Biological Oxygen Demand (BOD) and Chemical Oxygen Demand (COD) values recorded (Olafusi et al., 2018) confirm that the aquifers are receiving high loads of labile organic carbon, which can deplete dissolved oxygen, alter redox conditions, and subsequently facilitate the reductive dissolution of iron and manganese oxides, exacerbating the mobilization of sequestered heavy metals.

A critical analysis of the reviewed studies reveals that contamination severity is heavily modulated by local hydrogeological architecture. In the coastal sedimentary basins of Southern Nigeria (e.g., Lagos, Benin, Port Harcourt), characterized by high water tables (often < 5 m), shallow unconfined aquifers, and sandy/gravelly lithologies, leachate migration is rapid and largely unmitigated, explaining the extreme parameter values observed (Adeyemi et al., 2019; Wami et al., 2022). Conversely, in the crystalline basement complex of Northern Nigeria (e.g., Katsina, Kaduna, Kano), deeper water tables and the presence of weathered clayey overburdens may offer moderate natural filtration for certain cations (Musa et al., 2021). However, even in these regions, the ubiquitous presence of coliforms and elevated TDS indicates that these "protective" layers are insufficient against the sheer volume of leachate generated during the wet season, when infiltration rates peak.

From a public health perspective, this synthesis underscores a grim reality. The consumption of nitrate-laden water by infants is directly linked to methemoglobinemia (blue-baby syndrome), while elevated Pb exposure portends irreversible neurodevelopmental deficits in children. Concurrently, the chronic ingestion of hard water (exceeding 500 mg/L CaCO_3 as noted by Obaseki & Ikenweiwe, 2017) is epidemiologically associated with cardiovascular strain. When superimposed on the acute gastrointestinal distress from coliform contamination, the cumulative health burden on residents of these peri-urban settlements is substantial yet remains critically under-documented in morbidity statistics.

Despite the compelling evidence, the reviewed studies exhibit notable limitations. A striking methodological gap is the predominance of single-point, dry-season sampling campaigns. This cross-sectional approach fails to capture the seasonal fluctuation in contaminant plumes—whereas heavy rains accelerate vertical leaching, the dry season may concentrate ions via evaporation, creating a cyclically variable risk profile. Furthermore, while many studies

identify elevated parameters, few have employed geochemical speciation modeling (e.g., PHREEQC) or stable isotope tracers ($\delta^{15}\text{N}$, $\delta^{18}\text{O}$) to definitively differentiate anthropogenic leachate signatures from natural geogenic background enrichment. Such source-apportionment tools are imperative to prosecute evidence-based legal and policy decisions.

Finally, the regulatory landscape in Nigeria remains woefully lagging. While the NSDWQ provides benchmarks for finished drinking water, there is a conspicuous absence of binding federal legislation specifying minimum setback distances for cemeteries relative to boreholes—unlike the 50 to 100-meter statutory buffers recommended in many European jurisdictions (Ücisik & Rushbrook, 1998). This regulatory lacuna, compounded by the unplanned urbanization that frequently engulfs historical cemeteries, perpetuates a cycle of institutional negligence. The enforcement of the National Environmental Standards and Regulations Enforcement Agency (NESREA) is often hampered by logistical constraints and a lack of inter-agency coordination. Therefore, the transition from retrospective academic documentation to proactive, spatial planning-driven mitigation remains the paramount challenge.

In conclusion, this synthesis confirms that cemeteries are not merely spiritual resting places but active biogeochemical reactors that continuously challenge groundwater resilience. The confluence of elevated inorganic salts, toxic metals, and pathogenic microbes demands an urgent paradigm shift from passive acknowledgment to aggressive regulatory and engineering intervention, tailored to Nigeria's diverse hydrogeological realities. Future research must prioritize longitudinal monitoring, advanced geochemical fingerprinting, and pilot-scale phytoremediation trials to reclaim already compromised aquifers.

CONCLUSION

Groundwater from boreholes and wells in residential areas proximate to Nigerian cemeteries is critically vulnerable to physicochemical and microbial contamination. Decomposition by-products consistently alter the natural chemistry of aquifers, with Electrical Conductivity, Total Dissolved Solids, chloride, nitrate, sulphate, and heavy metals (particularly lead and iron) frequently exceeding NSDWQ and WHO safe thresholds. The pervasive presence of coliform bacteria further compounds the health burden, signalling faecal-organic contamination pathways that heighten the risk of waterborne diseases and chronic toxicities such as methemoglobinemia from nitrate exposure and neurodevelopmental deficits from lead ingestion.

The severity of contamination is not uniform but modulated by hydrogeological architecture—shallow, sandy aquifers in Southern Nigeria exhibit more rapid plume migration, while deeper, clay-rich overburdens in the North offer only partial attenuation. This variability, coupled with the predominance of single-point dry-season sampling in most studies, underscores critical gaps in both regulatory oversight and scientific methodology.

Consequently, this review advances the following evidence-based imperatives: first, NESREA and State Planning Boards must mandate hydrogeologically-differentiated setback zones— ≥ 100 metres for high-permeability soils and ≥ 50 metres for clay-rich terrains—to directly counter the documented proximity-dependent contamination; second, engineering controls such as impermeable liners and reactive interception trenches must be deployed to specifically target the identified ionic and heavy metal loads; third, public health surveillance must prioritize nitrate and coliform screening for

vulnerable populations within 200-metre buffer zones; fourth, regulatory agencies must enforce compulsory biennial wet- and dry-season groundwater monitoring to capture seasonal fluctuation and provide early warning; fifth, further forensic geochemical research utilizing stable isotope tracers is essential to definitively apportion contaminant sources and strengthen legal enforcement; and sixth, pilot phytoremediation trials should be funded around historical cemeteries in highly impacted urban centres to reverse the cumulative heavy metal loading already documented.

In sum, cemeteries are not merely spiritual resting places but active biogeochemical reactors that perpetually challenge groundwater resilience. Protecting public health and water security demands an urgent paradigm shift—from passive academic documentation to proactive, spatially-informed regulatory enforcement, integrated engineering intervention, and sustained community engagement. Without such decisive action, the cycle of contamination and preventable disease documented in this review will persist, disproportionately affecting Nigeria's most vulnerable peri-urban populations.

REFERENCES

- Adewumi, A. J., Oguntuase, A. M., & Adejuwon, J. O. (2021). Groundwater vulnerability to pollution from cemeteries: A review. *Journal of Environmental Management*, **287**, 112316. <https://doi.org/10.1016/j.jenvman.2021.112316>
- Adeyemi, O. O., Ojekunle, Z. O., & Taiwo, A. M. (2019). Impact of cemetery on groundwater quality in Ayobo, Lagos State, Nigeria. *Journal of Health and Pollution*, **9** (24), 191205. <https://doi.org/10.5696/2156-9614-9.24.191205>
- Akoto, O., Adiyiah, J., & Abrom, D. (2019). Physicochemical analysis of groundwater in selected communities in Nigeria. *Applied Water Science*, **9** (4), 86. <https://doi.org/10.1007/s13201-019-0968-z>
- Akinbile, C. O., Ogunrinde, T. A., & Olatunji, O. M. (2021). Groundwater contamination from a cemetery in Osun State, Nigeria: A case study. *Environmental Nanotechnology, Monitoring & Management*, **16**, 100505. <https://doi.org/10.1016/j.enmm.2021.100505>
- Bala, A., Sule, I. M., & Salami, A. T. (2020). Assessment of groundwater quality around a cemetery in Kano Metropolis, Nigeria. *Nigerian Journal of Environmental Sciences and Technology*, **4** (2), 432-443. <https://doi.org/10.36263/nijest.2020.02.0202>
- Da Silva, G. C., de Oliveira, I. B., & Jardim, F. A. (2021). Hydrogeochemical assessment of cemetery influence on groundwater quality in a tropical urban area. *Environmental Earth Sciences*, **80** (9), 345. <https://doi.org/10.1007/s12665-021-09636-0>
- Musa, J. J., Adewumi, J. R., & Ezenwa, I. M. (2021). Vulnerability assessment of groundwater contamination near a cemetery in Kaduna, Nigeria. *Arabian Journal of Geosciences*, **14** (5), 387. <https://doi.org/10.1007/s12517-021-06762-8>
- Nnadozie, P. C., Orij, J. C., & Umeh, S. I. (2020). Evaluation of the impact of cemetery on groundwater quality in Umuahia, Abia State, Nigeria. *International Journal of Environment and Pollution Research*, **8** (2), 1-15.
- Obaseki, M., & Ikenweiwe, N. B. (2017). Impact of Ikhueniro cemetery on groundwater quality in Benin City, Edo State, Nigeria. *Journal of Applied Sciences and Environmental Management*, **21** (7), 1283-1288. <https://doi.org/10.4314/jasem.v21i7.13>
- Olafusi, O. S., Oladejo, K. T., & Ayejuyo, O. O. (2018). Assessment of groundwater contamination near a cemetery in Yaba, Lagos State, Nigeria. *Journal of Chemical Society of Nigeria*, **43** (3), 502-509.
- Sangodoyin, A. Y., & Amori, A. A. (2019). Influence of cemetery on groundwater quality in Ibadan, Nigeria. *Environmental Technology & Innovation*, **14**, 100335. <https://doi.org/10.1016/j.eti.2019.100335>
- Ücisik, A. S., & Rushbrook, P. (1998). The impact of cemeteries on the environment and public health: An introductory briefing. World Health Organization Regional Office for Europe.
- Wami, E. N., Obenade, O., & Amakiri, A. O. C. (2022). Physicochemical and microbial analysis of water from wells close to a cemetery in Port Harcourt, Nigeria. *Journal of Water and Health*, **20** (1), 217-228. <https://doi.org/10.2166/wh.2021.168>
- World Health Organization. (2011). Guidelines for drinking-water quality (4th ed.).
- Żychowski, J. (2018). The impact of cemeteries on the natural environment. A review. *Journal of Ecology and Engineering*, **19** (5), 207-212. <https://doi.org/10.12911/22998993/91877>

